California Polytechnic State University **Economics 440**
Department of Economics **Spring 2026**

Advanced Econometrics:
Introduction to Causal Inference and Program Evaluation

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Class Time: TR from 2:10 pm to 4:00 pm
Class Place: 035-0217D
Office Hours: Thursdays from 12:30pm to 1:30pm and 4:15 pm to 6:00 pm or by appointment.

Course Description

The purpose of this course is to expose students to a variety of standard and advanced econometric techniques employed to uncover causal effects. The techniques to be studied are now widely used in economics, public policy, political science, and many other social sciences, as well as in non-academic settings in industry (e.g., tech, consulting, advertising), government, and non-governmental organizations (among many other places) to evaluate the effect of policies, programs, or interventions. We will use the potential outcome framework as a general principle to study identification and estimation of causal effects under various types of assumptions. While we start with randomized experiments (the gold standard for uncovering causal relationships), the main focus of the class is on methods aimed at addressing the challenging problem of answering causal questions in non-experimental settings. Time-permitting, we will cover the following non-experimental methods: covariate-adjustment methods (linear regression, matching, propensity-score methods), instrumental variables, regression discontinuity, and difference-in-differences. For each of the methods covered, the main focus of the class will be on understanding the intuition behind it, its key assumptions and limitations, as well as how to implement it in practice and interpret its results.

Prerequisites

Economics 339, or equivalent.

Textbook, Journal Articles, and Class Notes

It is hard to find a book that covers all the topics we will study in this course at an undergraduate level. The closest such book to date is the one by Angrist and Pischke (2014):

> Angrist and Pischke (2014). *Mastering 'Metrics: The Path from Cause to Effect*. Princeton University Press.

While this class will *not* be entirely based on this textbook, it does contain most of the main ideas and concepts that I want you to learn in this class. Hence, <u>I will be assigning readings from this book as we move along.</u> However, not all the topics that I will discuss in class can be found in this book (e.g., matching). To complement the book, I will assign readings from journal articles and other sources, all of which will be available on Canvas. In addition, typed notes of *some* of the material covered in the lectures will be provided via Canvas. These notes will include some text (e.g., "bullet points" and key concepts/theorems) and some equations; however, they will not be complete or self-contained as some of the equations as well as all the pictures and graphs will be worked out during lectures. The problem sets and exams will be based on the material covered in class (hence, it is very important that you regularly attend/review lectures).

Some not-too-technical survey articles and books that you may find useful (to be available on Canvas), which contain some of the topics to be covered in class, are:

1. Caliendo, M. and R. Hujer (2005). "The Microeconometric Estimation of Treatment Effects – An Overview," IZA DP No. 1653.
2. Schlotter, M., Schwerdt, G. and L. Woessmann (2010). "Econometric Methods for Causal Evaluation of Education Policies and Practices: A Non-Technical Guide," IZA DP No. 4275.
3. Khandker, S., Koolwal, G. and H. Samad (2010). *Handbook on Impact Evaluation: Quantitative Methods and Practices*. World Bank Publications.
4. Gertler, P., Martinez, S. Premand, P., Rawlings, L. and Vermeersch C. (2011). *Impact Evaluation in Practice*. World Bank Publications.

Some more advanced (and technical) survey articles that you may find useful to learn more about the methods to be discussed (some to be available on Canvas):

1. Imbens, G. and J. Wooldridge (2009), "Recent Developments in the Econometrics of Program Evaluation", *Journal of Economic Literature*, 47 (1), pp. 5-86.
2. Abadie, A. and M. Cattaneo (2018). "Econometric Methods for Program Evaluation," *Annual Review of Economics*, 10 (1), pp. 465-503.
3. Angrist, J. and A. Krueger (1999). "Empirical Strategies in Labor Economics." In Orley Ashenfelter and David Card, eds., *Handbook of Labor Economics*, Volume 3A. Amsterdam: North-Holland, pp. 1277-1366.
4. DiNardo, J. and D. Lee (2010). "Program Evaluation and Research Designs," NBER Working Paper 16016.
5. Heckman, J., R. LaLonde and J. Smith (1999). "The Economics and Econometrics of Active Labor Market Programs." In Orley Ashenfelter and David Card, eds., *Handbook of Labor Economics*, Volume 3A. Amsterdam: North-Holland, pp. 1865-2097.
6. Blundell, R. and M. Costa Dias (2009), "Alternative Approaches to Evaluation in Empirical Microeconomics", *Journal of Human Resources*, 44 (3), pp. 565-640.
7. Angrist and Pischke (2009). *Mostly Harmless Econometrics*. Princeton University Press.

Course Website

The course website will be available through Canvas. I will post on this website some of the material for the course, including additional readings, problem sets, assignments, and solutions.

Homework

There will be approximately four problem sets involving analysis of real data sets to develop the ability to apply the methods discussed in class. You may use Matlab, R, Stata, or any other software for your computations (I will use Stata to solve the problem sets, so the answer keys will be given using this program; more on this below). ***All problem sets are to be submitted via Canvas by 11:59 pm of the day it is due***. To the extent possible, please submit each of your problem sets as a **single PDF file**. If that is not possible, please try to minimize the number of files you submit. If everybody submits multiple files, I could end up with a couple hundred files, which gets harder to manage. Also, note that PDF files are strongly preferred. If you need to scan something (e.g., handwritten answers) to put it in PDF format and do not have a scanner, you may use a mobile app such as CamScanner (https://www.camscanner.com). In each problem set, **you will be required to turn in a stand-alone "master computer program file" (e.g. a do-file in Stata) that can be run to (easily) reproduce your results.** No points will be awarded to answers for which the commands used to produce the results are not clearly marked in the program file. As mentioned above, it will be great if you could convert that file to PDF and attach it to the rest of your problem set so you submit a single PDF file. **NO late submissions will be accepted.** Finally, after each problem set is due, I will provide its answer key, which will include detailed explanations for how to solve each of the questions in it.

Computer Software

The recommended software for the assignments is **Stata**. This is an intuitive, powerful, and widely-used econometric software. Using Stata will simplify your experience in this class considerably since, while you are free to use any software of your choice (e.g., SAS, Matlab, Gauss, R, etc.), you will be responsible for any technical support you may need to complete the assignments in other software languages. In any event, keep in mind that, even if using Stata, you are ultimately responsible for the completion of your own assignments.

Stata is available to students on the OCOB computer labs (at least this was the case in the past). You may also buy a 6-month license of the "Basic Edition" version of Stata or "Stata/BE" (which is enough for this class) for about $50 here:
https://www.stata.com/order/new/edu/gradplans/student-pricing/.

If you are not familiar with Stata, there are several great online tutorials on how to use it (see, e.g., https://www.stata.com/links/resources-for-learning-stata/). If you are completely new to Stata, a good place to start is: https://www.princeton.edu/~otorres/Stata/.

Summary of Journal Articles (or "Write-ups")

Throughout the course you will be asked to read and briefly discuss between four to five journal articles relevant to the topics we will discuss in class. Further guidelines regarding these summaries will be provided on Canvas. The purpose of these write-ups is twofold. First, some of the topics we will cover are not available in undergraduate econometrics books, so we will go back to the original articles written on the topic. Some of these papers will be used for assignments, too. Second, and more general, these assignments will expose you to reading papers from top economics and statistics journals. When you leave school, many times you will have to read about methods that were not covered in school (it is impossible to cover everything!) or were developed after you left. Reading those articles will be a good exercise to prepare you for that.

Midterms

There will be one midterm exam. The *tentative* date for the midterm is: Thursday May 7th.
There will be **NO** make-up midterms. If a student misses a midterm because of a valid excuse (e.g., medical reasons), its weight will be put into the final. The midterm is **not** optional.

Final Exam

The final exam will cover the material *from the entire course*. The date for the final exam is determined by Cal Poly and cannot be changed. The final will be on Tuesday June 9th from 4:10 pm to 7:00 pm. There will be **NO** make-up final.

Grading

The maximum number of points you can obtain is 100. The distribution of points is as follows:

Homework	20
Summary of Journal Articles	10
Midterm	35
Final Exam	35
Total:	100

Academic Ethics

Cal Poly does not tolerate academic dishonesty and penalizes cheating or plagiarizing (interpreting as one's own anything done by another) on assignments and exams. Please visit https://academicprograms.calpoly.edu/content/academicpolicies/Cheating. Bottom line: **please don't cheat**.

Final Comment:

This will be a demanding course that will require a lot of work from your part in terms of readings and empirical exercises (in addition to the difficulty coming from the technical nature of econometrics). I will work hard to explain you in a clear way the techniques and concepts I expect you to learn, and to provide you with useful problem sets and readings to help you grasp the class material. But like in every other class, it is your responsibility to take control of your own education. If you are having problems, I am more than willing to help; you just need to approach me at some point. Therefore, please feel free to ask questions during our class, and if you need help with a particular concept, technique, or exercise, you are welcome to stop by my office hours or to schedule an appointment. I hope you learn a lot of useful econometric tools from this course!

(Tentative) Outline of Topics

1. Introduction and Review of Probability and Statistics
2. The Potential Outcome Framework
3. Randomized Control Trials
4. Covariance-Adjustment Methods
    a. Linear Regression
    b. Matching
    c. Methods based on the Propensity Score
5. Instrumental Variables
6. Regression Discontinuity*
7. Difference-in-Differences and Two-Way Fixed Effect Methods*
8. Synthetic Control Methods*
9. External and Internal Validity*
10. Bounds for Treatment Effects*
11. Causal Mediation Analysis*

(* Time Permitting. These topics may or may not be covered. Indeed, this is an extensive list of topics to be covered in one quarter. Most likely we will not have time to cover these topics. I list them here in case you are interested and want to learn more about Causal Inference and Program Evaluation.)